Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-07-25

Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pendi

Profile snapshot

- **Primary site:** bone marrow (hematologic)
- **Histology:** Acute myeloid leukemia, myelodysplasia-related (AML-MR), relapsed/refractory; transformed from prior MDS (RCMD, IPSS-R high/very-high, diagnosed ~2019); with extramedullary myeloid sarcoma
- **Stage:** Relapsed/refractory AML-MR. Marrow ~85% blasts (June 2026); peripheral-blood blasts 58->82% (late June 2026); extramedullary right-thigh myeloid sarcoma on MRI (May 2026, adverse feature). Refractory to FLAG-IDA + venetoclax reinduction. Late relapse ~6 years after matched-sibling PBSC allo-HCT (~2020); pattern consistent with graft-versus-leukemia escape rather than graft failure.
- **ECOG:** 1
- **Age band:** 50-59
- **Sex:** female
- **Biomarkers:** TP53 aberrant, pending confirmation (NOT resolved in either direction). Pathogenic TP53 mutation reported on 2019 diagnostic NGS; 2026 relapse NGS reported NEGATIVE, but on a limited assay (LOD ~2% VAF, no copy-number/CNV detection, not MRD-grade). Discrepancy most plausibly subclonal loss, copy-loss, or a re-review artifact rather than true reversion. Working call: TP53-aberrant pending confirmation. (ngs_pending); TP53 Y220C (specific variant) unknown / to be determined — Y220C status not established; therapy handle (rezatapopt) is contingent on it (ngs_pending); HLA-A*02:01 unknown / pending retrieval. High-resolution HLA typing was performed ~2020 but the allele-level result is not in the records on hand (placeholder in the EHR). (unknown); HLA-A*24:02 unknown / pending (unknown); HA-1 / HA-2 minor histocompatibility antigens (recipient and sibling donor) unknown / pending for BOTH patient and sibling donor. HLA-identical siblings can still differ at HA-1/HA-2. (unknown); CD33 positive (blasts CD33+ by flow cytometry); CD123 positive (blasts CD123+ by flow cytometry); AML immunophenotype (supporting) CD34+ (increased), CD13+, CD15+ (subset), CD117+ (variably increased), HLA-DR (variably decreased), CD38 decreased; negative CD4/CD5/CD7/CD14/CD16/CD19/CD56/CD64; FLT3 negative / no mutation (2019 diagnostic NGS); IDH1 negative / no mutation (2019 diagnostic NGS); IDH2 negative / no mutation (2019 diagnostic NGS); JAK3 variant of uncertain significance (VUS) — 2019 (unknown); SMC3 variant of uncertain significance (VUS) — 2019 (cohesin-complex gene) (unknown); Cytogenetics / karyotype Complex, adverse-risk. 2019: complex karyotype with del(5q), del(7q), del(20q12), der(17)t(13;17) [17p], monosomy 13, others. 2026 relapse: gain of 5' MECOM, del(5q), del(7q), KMT2A amplification (4 copies), del(8cen), del(16q22). Shared del(5q)/del(7q) + TP53 founder across timepoints indicates evolution of the ORIGINAL clone transformed to AML (not a new/secondary or donor-derived leukemia). (ngs_pending)
- **Efficacy/toxicity weight:** 0.75

20 ranked therapeutic options across 7 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Hct2 Platform interventions

1. Iomab-B (131I-apamistamab) anti-CD45 radioimmunotherapy conditioning + second allo-HCT

Likelihood of effect: Moderate: the only randomized signal in the dossier (SIERRA dCR 17.1% vs 0%, p<0.0001), read indirectly onto a prior-HCT relapse it did not enroll.

Toxicity burden: High (grade 3+ TRAEs 59.7%, at parity with conventional care — conditioning cytopenias, infection, mucositis; second-transplant TRM uncharacterized)

Counter-productive MoA:Low (Radioconjugate marrow injury could compromise engraftment if the HCT2 platform is not secured; no mechanism objection raised.)

Overall:The curative backbone all five personas endorsed — the randomized way to carry active disease into a second transplant, gated on a fitness workup that has not yet been done.

Key references: PMID 39298738 · NCT02665065 · NCT07157514

Evidence in detail

Iomab-B (131I-apamistamab) anti-CD45 radioimmunotherapy conditioning + second allo-HCT

Gyurkocza et al. *J Clin Oncol* 2025

- **Disease context:Active relapsed/refractory AML, age >=55, planned allo-HCT** (2L+) — Active R/R AML age >=55 fit for allo-HCT; ITT included crossover (52% of conventional-care arm received 131I-apamistamab). Enrolled patients heading to a first allo-HCT.
- **n:** 153
- **Toxicities:Any treatment-related AE** (grade >=3): 59.7% (n=76)
- **Efficacy:CR_rate** 17.1% (durable CR >=180d) (95% CI 9.4-27.5); **HR_OS** 0.99 (95% CI 0.7-1.41); **HR_EFS** 0.23 (95% CI 0.15-0.34)

Hla A0201 interventions

1. HA-1-specific CD8+/CD4+ TCR-T (Bleakley platform; TSC-100 / BSB-1001) *(conditional on hla_a0201 positive)*

Likelihood of effect: Assuming A*02:01 + HA-1 mismatch confirm: uncertain but curative-intent — 4/9 CR in Krakow 2024 (n=9 feasibility, no comparator). Negative typing forecloses it.

Toxicity burden: Low (no CRS, no ICANS, no DLTs across n=9; lymphodepletion neutropenia and grade 3-5 infection; mild grade 1-2 GVHD)

Counter-productive MoA:Low (TCR-T kinetics may lag an 85%-blast marrow and antigen-loss escape is possible; lineage-restricted killing limits off-target collateral.)

Overall:The only lever that answers the referral's own question — a clean-safety, curative-intent graft-versus-leukemia consolidation, live only if the pending typing confirms and delivered after HCT2.

Key references: PMID 38683966 · PMID 29051183 · NCT03326921

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

CBX-250 (CG1/HLA-A*02:01 TCR-mimetic T-cell engager) — _Thin evidence_

Likelihood of effect: Unknown: first-in-human phase 1, no efficacy readout; the argument is the off-the-shelf no-lead-time timeline, not a proven number.

Toxicity burden: Moderate (T-cell-engager cytokine-release risk against high blast burden; no reported grade profile yet)

Counter-productive MoA:Low (Antigen escape or T-cell exhaustion possible; no board mechanism objection, and no efficacy data to weigh either way.)

Overall:An off-the-shelf A*02:01 engager with a real no-lead-time advantage and 11 recruiting US sites, but zero efficacy data — enrollable yet unproven, gated on the same allele call.

Key references: NCT06994676

TSC-100 (donor-derived HA-1-specific TCR-T) — _Consolidated_

Likelihood of effect: Consolidated into rank 3 (ha1-tcr-t): the ranked HA-1 approach's likelihood applies; no separate efficacy data for the commercial product.

Toxicity burden: Low (no clinical data yet; class safety read carried by the Bleakley HA-1 platform — 0 CRS, 0 ICANS)

Counter-productive MoA:Low (As for the ranked HA-1 approach: antigen-loss escape possible, lineage-restricted killing limits collateral.)

Overall:The commercial HA-1 arm of the ranked approach, recruiting at 21 US sites — same gate, same board signal, consolidated under rank 3.

Key references: NCT05473910 · PMID 38683966

TSC-101 (donor-derived HA-2-specific TCR-T) — _Consolidated_

Likelihood of effect: Consolidated into rank 3 (ha1-tcr-t): applies if the mismatch is at HA-2; no separate efficacy data.

Toxicity burden: Low (no clinical data yet; class safety read carried by the HA-1 platform)

Counter-productive MoA:Low (As for the ranked HA approach: antigen-loss escape possible, lineage-restricted killing limits collateral.)

Overall:The HA-2 arm of the ranked approach, opening if the pair mismatches at HA-2 instead of HA-1 — consolidated under rank 3.

Key references: NCT05473910

BSB-1001 (HA-1-directed TCR-T) — _Consolidated_

Likelihood of effect: Consolidated into rank 3 (ha1-tcr-t): the ranked HA-1 approach's likelihood applies; no separate efficacy data.

Toxicity burden: Low (no clinical data yet; class safety read carried by the HA-1 platform)

Counter-productive MoA:Low (As for the ranked HA-1 approach: antigen-loss escape possible, lineage-restricted killing limits collateral.)

Overall:A newer HA-1 TCR-T that widens site access for the ranked approach — same gate, same board signal, consolidated under rank 3.

Key references: NCT06704152

Evidence in detail

HA-1-specific CD8+/CD4+ TCR-T (Bleakley platform; TSC-100 / BSB-1001)

Krakow et al. *Blood* 2024

- **Disease context:Recurrent leukemia/myeloid neoplasm after allo-HCT** (2L+) — Recurrent AML/ALL/MDS/JMML after allo-HCT; HLA-A*02:01-positive with recipient HA-1(H)-positive / donor HA-1-negative mismatch. n=9 treated.
- **n:** 9
- **Toxicities:CRS** (grade any): 0% (n=9); **ICANS** (grade any): 0% (n=9); **GVHD (acute)** (grade 1-2): (n=9); **Neutropenia** (grade >=3): (n=9); **Infection** (grade >=3): (n=9)
- **Efficacy:CR_rate** 4/9 patients; **AE_rate** 0 DLTs

TSC-100 (donor-derived HA-1-specific TCR-T)

Krakow et al. *Blood* 2024

- **Disease context:Recurrent leukemia/myeloid neoplasm after allo-HCT** (2L+) — Recurrent AML/ALL/MDS/JMML after allo-HCT; HLA-A*02:01-positive with recipient HA-1(H)-positive / donor HA-1-negative mismatch. n=9 treated.
- **n:** 9
- **Toxicities:CRS** (grade any): 0% (n=9); **ICANS** (grade any): 0% (n=9); **GVHD (acute)** (grade 1-2): (n=9); **Neutropenia** (grade >=3): (n=9); **Infection** (grade >=3): (n=9)
- **Efficacy:CR_rate** 4/9 patients; **AE_rate** 0 DLTs

Cd123 interventions

1. Tagraxofusp (SL-401, CD123-directed IL-3/diphtheria-toxin fusion)

Likelihood of effect: Low-to-uncertain for AML: headline 72% CR is first-line BPDCN cross-tumor (Pemmaraju 2019); no AML efficacy data and CD123 density still pending.

Toxicity burden: High (capillary leak syndrome with treatment-related deaths, transaminitis ALT 64% / AST 60%, hypoalbuminemia 55%)

Counter-productive MoA:Low (No mechanistic antagonism; the efficacy risk is cross-tumor non-transfer from BPDCN, and on-target killing of normal CD34+CD123+ progenitors adds myelosuppression.)

Overall:The only labeled CD123 agent with a real safety algorithm, but its headline efficacy is BPDCN cross-tumor and its capillary-leak deaths demand organ-function labs first.

Key references: PMID 31018069 · NCT06561152 · PMID 29773600

2. Pivekimab sunirine (IMGN632, CD123 ADC)

Likelihood of effect: Moderate as debulking: own R/R AML data (ORR 21%, 95% CI 8-40; cCR 17%), but single-agent depth is modest against an 85%-blast marrow.

Toxicity burden: High (veno-occlusive disease at higher doses, one treatment-related death; RP2D febrile neutropenia 10%, infusion reactions 7%)

Counter-productive MoA:High (Veno-occlusive disease from the ADC ahead of HCT2 could sink the very transplant the debulking is meant to enable.)

Overall:The CD123 ADC with the best own-disease data, held under a live, conditional VOD veto — it reinstates as a debulking bridge the moment hepatic and albumin baselines clear it.

Key references: PMID 38423051 · PMID 29661755 · NCT03386513

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

CLL-1/CD33/CD123-directed CAR-T — _Not enrollable_

Likelihood of effect: Not practically enrollable (China single-site); dual CD33/CD123 targeting hedges escape but no accessible route.

Toxicity burden: Moderate (CAR-T cytokine-release and lymphodepletion risk; no reported grade profile)

Counter-productive MoA:Low (Multi-antigen design limits single-antigen escape; on-target myelosuppression from CD33/CD123 CAR-T possible.)

Overall:The dual CD33/CD123 CAR-T the profile's escape-hedging logic calls for, but China-only single-site — informational, not a reachable route against US options.

Key references: NCT04010877

CD123-CD33 compound CAR-T (cCAR) — _Not enrollable_

Likelihood of effect: Not practically enrollable (China sites, stale/unknown status); dual-antigen match but no verified accessible route.

Toxicity burden: Moderate (CAR-T cytokine-release and lymphodepletion risk; no reported grade profile)

Counter-productive MoA:Low (Compound design limits single-antigen escape; on-target myelosuppression possible.)

Overall:A dual CD123/CD33 compound CAR-T matching the antigen profile, but the registry status is stale and sites are China-only — status must be confirmed before it counts as reachable.

Key references: NCT04156256

Flotetuzumab (MGD006, CD123xCD3 DART) — _Unavailable_

Likelihood of effect: Not enrollable (program discontinued): 30% ORR in refractory AML (Uy 2021), TP53 signal from a post-hoc correlative subgroup (Vadakekolathu 2020).

Toxicity burden: Moderate (cytokine release / infusion reactions, largely grade 1-2 with step-up dosing, dexamethasone, tocilizumab)

Counter-productive MoA:Moderate (CD123xCD3 redirection needs fit autologous T cells; exhaustion after transplant and FLAG-IDA may blunt it; TP53 rationale is post-hoc.)

Overall:A discontinued drug carrying a live mechanism signal — its TP53-inflamed-marrow rationale is post-hoc, and the reason to chase an open CD123xCD3 construct rather than this one.

Key references: PMID 32929488 · PMID 33057635 · NCT02152956

Vibecotamab (XmAb14045, CD123xCD3 bispecific) — _Unavailable_

Likelihood of effect: Not enrollable (discontinued, MRD-setting mismatch).

Toxicity burden: Moderate (CD123xCD3 cytokine-release class risk; no case-relevant data)

Counter-productive MoA:Low (Same T-cell-engager class; no goal-specific antagonism, no board mechanism objection.)

Overall:A discontinued CD123xCD3 bispecific studied in the wrong disease state — informational pipeline context only.

Key references: NCT05285813 · PMID 37647601

Evidence in detail

Tagraxofusp (SL-401, CD123-directed IL-3/diphtheria-toxin fusion)

Pemmaraju et al. *N Engl J Med* 2019

- **Disease context:Blastic plasmacytoid dendritic-cell neoplasm (BPDCN), CD123+** (any) — CD123+ BPDCN; 29 previously untreated at 12 μg/kg (primary-outcome cohort) plus 15 previously treated. Not the patient's disease.
- **n:** 47
- **Toxicities:Capillary leak syndrome** (grade any): 19% (n=47); **ALT increased** (grade any): 64% (n=47); **AST**

increased (grade any): 60% (n=47); **Hypoalbuminemia** (grade any): 55% (n=47); **Peripheral edema** (grade any): 51% (n=47); **Thrombocytopenia** (grade any): 49% (n=47)
- **Efficacy:CR_rate** 72%; **ORR** 90%; **OS_rate** 59% at 18 mo; **OS_rate** 52% at 24 mo; **ORR** 67%; **OS** 8.5 months

Pivekimab sunirine (IMGN632, CD123 ADC)

Daver et al. *Lancet Oncol* 2024

- **Disease context:Relapsed/refractory AML, CD123+** (2L+) — CD123+ R/R AML, ECOG 0-1; first-in-human dose-escalation/expansion, RP2D expansion cohort n=29 (whole study n=91).
- **n:** 91
- **Toxicities:Febrile neutropenia** (grade >=3): 10% (n=29); **Infusion-related reaction** (grade >=3): 7% (n=29); **Anaemia** (grade >=3): 7% (n=29); **Veno-occlusive disease** (grade >=3): (n=68); **Treatment-related death** (grade 5): (n=68)
- **Efficacy:ORR** 21% (95% CI 8-40); **CR_rate** 17% (composite CR) (95% CI 6-36)

Flotetuzumab (MGD006, CD123xCD3 DART)

Uy et al. *Blood* 2021

- **Disease context:Primary induction failure / early-relapse R/R AML** (2L+) — Primary induction failure or early relapse (<6 mo) R/R AML; benefit enriched in an immune-infiltrated tumor microenvironment. RP2D PIF/ER subset n=30 (whole study n=88).
- **n:** 88
- **Toxicities:CRS / infusion-related reaction** (grade any): (n=88); **CRS / infusion-related reaction** (grade >=3): (n=88)
- **Efficacy:CR_rate** 26.7% (CR/CRh); **ORR** 30%; **OS** 10.2 months; **OS_rate** 75% at 6 mo; **OS_rate** 50% at 12 mo

Vibecotamab (XmAb14045, CD123xCD3 bispecific)

***Blood Adv* 2023**

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Wt1 interventions

1. Multiantigen leukemia-specific donor T cells (PRAME/WT1/Survivin/NY-ESO-1)

Likelihood of effect: Low-to-modest: 2/8 objective responses in active disease (Lulla 2021); hits WT1/PRAME without a single-HLA gate, but WT1/PRAME expression unconfirmed.

Toxicity burden: Low (no grade >2 acute GVHD, no extensive chronic GVHD across n=25)

Counter-productive MoA:Low (Multi-antigen targeting limits single-antigen escape; no mechanism that blunts the goal.)

Overall:A multi-antigen donor T-cell hedge that sidesteps the A*02:01 gate and is enrollable post-transplant — but published activity is modest and WT1/PRAME expression is still to confirm.

Key references: PMID 33270816 · NCT02203903

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

FH-WT1-E50 (anti-WT1 TCR/CD8ab T cells) + azacitidine — _Not enrollable_

Likelihood of effect: Not currently enrollable (MRD-setting mismatch); WT1 A*02:01 TCR-T precedent is prophylactic (Chapuis 2019), not active-disease salvage.

Toxicity burden: Low (WT1 TCR-T class; no active-disease safety data for this construct)

Counter-productive MoA:Low (WT1 is broadly expressed on normal progenitors; on-target myelosuppression possible, no goal-blunting vector.)

Overall:The open WT1 A*02:01 TCR-T, but it enrolls MRD-positive disease — a consolidation-phase option for after cytoreduction, not a current-state fit.

Key references: NCT07645469 · PMID 31235963

NTLA-5001 (CRISPR-engineered WT1-directed TCR-T) — _Unavailable_

Likelihood of effect: Not enrollable (terminated program).

Toxicity burden: Low (no case-relevant data; program halted)

Counter-productive MoA:Low (WT1 TCR-T class; no goal-specific antagonism.)

Overall:A terminated WT1 TCR-T program — closes the loop on the WT1 pipeline, no route.

Key references: NCT05066165

WT1-TCRC4 EBV-specific CD8+ Tcm (Chapuis/Greenberg) — _Unavailable_

Likelihood of effect: Not enrollable (completed); precedent value only — prophylactic RFS 100% at 44 mo (Chapuis 2019).

Toxicity burden: Low (EBV-specific CD8 Tcm design to minimize GVHD; no treatment-limiting signal reported)

Counter-productive MoA:Low (WT1 TCR-T class; no goal-specific antagonism.)

Overall:The WT1 A*02:01 TCR-T precedent (prophylaxis, RFS 100% at 44 mo) — completed and succeeded by FH-WT1-E50, no open route to this construct.

Key references: PMID 31235963 · NCT01640301

WT1-siTCR gene-transduced lymphocytes (TBI-1301; HLA-A*24:02-restricted) — _Unavailable_

Likelihood of effect: Not enrollable today (no open A*24:02 WT1-siTCR successor); route-preservation only — transient blast reductions (2/8, Tawara 2017).

Toxicity burden: Low (no normal-tissue toxicity reported across n=8)

Counter-productive MoA:Low (WT1 siTCR silences endogenous TCR to limit mispairing; no goal-blunting vector, spares normal progenitors preclinically.)

Overall:The A*24:02 WT1 fallback that keeps an immunotherapy route alive if A*02:01 is negative — but no open trial today, so it is a reason to type both alleles, not a current route.

Key references: PMID 28860210 · PMID 21673345

Evidence in detail

Multiantigen leukemia-specific donor T cells (PRAME/WT1/Survivin/NY-ESO-1)

Lulla et al. *Blood* 2021

- **Disease context:AML/MDS after allo-HCT (prophylaxis and treatment)** (2L+) — Post-HCT AML/MDS; 17 at high risk of relapse (prophylaxis) and 8 with active disease; products generated from 29 donors, 25 patients infused.
- **n:** 25
- **Toxicities:GVHD (acute)** (grade >=3): 0% (n=25); **GVHD (chronic, extensive)** (grade any): 0% (n=25)
- **Efficacy:ORR** 2/8 patients; **OS** NR months; **PFS** NR months (leukemia-free survival)

FH-WT1-E50 (anti-WT1 TCR/CD8ab T cells) + azacitidine

Chapuis et al. *Nat Med* 2019

- **Disease context:High-risk AML, post-HCT relapse prophylaxis** (maintenance) — HLA-A*02:01-positive, WT1-positive high-risk AML given as post-HCT relapse prophylaxis; compared with a concurrent 88-patient similar-risk cohort.
- **n:** 12
- **Toxicities:GVHD** (grade any): (n=12)
- **Efficacy:RFS** 100% at median 44 mo; **RFS** 54%

WT1-TCRC4 EBV-specific CD8+ Tcm (Chapuis/Greenberg)

Chapuis et al. *Nat Med* 2019

- **Disease context:High-risk AML, post-HCT relapse prophylaxis** (maintenance) — HLA-A*02:01-positive, WT1-positive high-risk AML given as post-HCT relapse prophylaxis; compared with a concurrent 88-patient similar-risk cohort.
- **n:** 12
- **Toxicities:GVHD** (grade any): (n=12)
- **Efficacy:RFS** 100% at median 44 mo; **RFS** 54%

WT1-siTCR gene-transduced lymphocytes (TBI-1301; HLA-A*24:02-restricted)

Tawara et al. *Blood* 2017

- **Disease context:Refractory AML and high-risk MDS** (2L+) — HLA-A*24:02-positive, WT1-positive refractory AML and high-risk MDS; two dose cohorts, n=8.
- **n:** 8
- **Toxicities:On-target off-tumor (normal tissue) toxicity** (grade any): 0% (n=8)
- **Efficacy:other** 2/8 patients; **OS_rate** 4/5 patients >12 mo

Tp53 Y220C interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Rezatapopt (PC14586, Y220C-selective p53 reactivator) + azacitidine — _Thin evidence_

Likelihood of effect: Unknown clinically: strong Y220C AML preclinical signal (Carter 2025) but no peer-reviewed clinical efficacy, and killing required venetoclax the disease has progressed through.

Toxicity burden: Low (none extractable — no published clinical toxicity yet)

Counter-productive MoA:Low (Single-agent reactivation is largely cytostatic without BCL-2 inhibition; prior venetoclax progression may blunt the combination that makes it work.)

Overall:The Y220C-selective TP53 handle that enters contention only if the variant confirms — double-gated on TP53 status and Y220C, with no clinical efficacy data yet to rank on.

Key references: PMID 40608889 · PMID 39945593 · NCT06616636

Prame interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

BPX-701 (PRAME TCR-T with iCasp9 safety switch) + rimiducid — _Unavailable_

Likelihood of effect: Not enrollable (terminated program, sponsor wound down).

Toxicity burden: Low (no case-relevant data; program halted)

Counter-productive MoA:Low (PRAME TCR-T class; no goal-specific antagonism.)

Overall:A terminated PRAME TCR-T — the open PRAME route is the multiantigen TAA-T product at rank 5.

Key references: NCT02743611

Tp53 interventions

Also considered — not ranked

Feature-targeting investigational options the board considered but did not rank as live top-tier choices: Unavailable (program discontinued), Consolidated (one product of an approach ranked above), Thin evidence (no peer-reviewed clinical efficacy yet), or Not enrollable (cross-tumor or geographically inaccessible).

Eprenetapopt (APR-246, pan-p53 reactivator) + venetoclax + azacitidine — _Unavailable_

Likelihood of effect: Low: pan-reactivator ORR 64-71% single-arm, but the randomized TP53-mutant MDS phase 3 missed its endpoint; not Y220C-selective.

Toxicity burden: High (febrile neutropenia 47%, thrombocytopenia 37%, leukopenia 25%, one treatment-related death)

Counter-productive MoA:Low (Pan-reactivation is non-selective; the low-yield concern is efficacy failure, not active antagonism.)

Overall:The pan-p53 reactivator the board flags as low-yield — a negative randomized phase 3 and no Y220C selectivity; rezatapopt is the better TP53 track if Y220C confirms.

Key references: PMID 36990622 · PMID 33449813

Evidence in detail

Eprenetapopt (APR-246, pan-p53 reactivator) + venetoclax + azacitidine

Garcia-Manero et al. *Lancet Haematol* 2023

- **Disease context:TP53-mutant AML, treatment-naive** (1L) — Treatment-naive TP53-mutant AML with >=1 pathogenic TP53 mutation, ECOG 0-2; triplet expansion evaluable n=39 (whole study n=49). Pan-p53 reactivator, not Y220C-selective.
- **n:** 49
- **Toxicities:Febrile neutropenia** (grade >=3): 47% (n=49); **Thrombocytopenia** (grade >=3): 37% (n=49); **Leukopenia** (grade >=3): 25% (n=49); **Anaemia** (grade >=3): 22% (n=49); **Treatment-related death** (grade 5): 2% (n=49)
- **Efficacy:ORR** 64% (95% CI 47-79); **CR_rate** 38% (95% CI 23-55)

Sallman et al. *J Clin Oncol* 2021

- **Disease context:TP53-mutant MDS or oligoblastic AML** (1L) — TP53-mutant MDS (n=40), oligoblastic AML with 20-30% blasts (n=11), and MDS/MPN (n=4); frontline. 35% went on to allo-HCT.
- **n:** 55
- **Toxicities:Febrile neutropenia** (grade >=3): 33% (n=55); **Leukopenia** (grade >=3): 29% (n=55); **Neutropenia** (grade >=3): 29% (n=55)

- **Efficacy:ORR** 71%; **CR_rate** 44%; **CR_rate** 50%; **OS** 10.8 months